Expertise Answer and Mark Scheme Book

6 matched answers out of 214 expertise questions

Missing pages are intentional placeholders for later correction/source import.

Answer 1: Nov 2024 P2H Q21 **4 marks**

Topic: Set Notation & Venn Diagrams Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 2: Jun 2022 P1HR Q20 **4 marks**

Topic: Percentages Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 3: Jun 2025 P1HR Q26 **4 marks**

Topic: Percentages Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 4: Jan 2022 P1HR Q22 **4 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 5: Jun 2023 P2H Q25 **4 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 6: Jun 2025 P2H Q22 **4 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 7: May 2021 P1H Q20 **3 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 8: Nov 2024 P1H Q21 **3 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 9: Nov 2025 P2H Q24 **5 marks**

Topic: Rounding, Estimation & Bounds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 10: Jun 2022 P1HR Q21 **4 marks**

Topic: Surds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 11: May 2023 P1HR Q23 **4 marks**

Topic: Surds Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 12: Jan 2020 P2H Q23 3 marks

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 13: Jan 2022 P2H Q24 **5 marks**

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 14: Jun 2022 P2H Q21 **5 marks**

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 15: Jun 2025 P2H Q23 **4 marks**

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 16: Nov 2023 P2H Q22 **5 marks**

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 17: Nov 2025 P1H Q23 3 marks

Topic: Ratio Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 18: Jan 2022 P2HR Q20 **4 marks**

Topic: Direct & Inverse Proportion Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 19: Jun 2024 P2H Q20 **3 marks**

Topic: Direct & Inverse Proportion Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 20: Jan 2020 P2HR Q21 **7 marks**

Topic: Algebraic Roots & Indices Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 21: Jun 2022 P2H Q24 **3 marks**

Topic: Algebraic Roots & Indices Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 22: Jun 2024 P2H Q23 **3 marks**

Topic: Algebraic Roots & Indices Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 23: May 2023 P1HR Q24 **5 marks**

Topic: Algebraic Roots & Indices Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 24: May-Nov 2020 P1HR Q21 **5 marks**

Topic: Algebraic Roots & Indices Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 25: Jan 2020 P2HR Q23 3 marks

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 26: Jan 2022 P1H Q20 **4 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 27: Jun 2022 P1H Q24 **4 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 28: Jun 2022 P1HR Q23 **5 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 29: Jun 2025 P2H Q25 **6 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 30: May 2021 P2H Q22 **4 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 31: May 2024 P1H Q25 4 marks

Topic: Completing the Square Status: matched Source pages: 28

25	$\pm 3\left(x^2 \pm 4x\right)$..... or $\pm 3\left(x^2 \pm 4x.....\right)$ or $b = 3$		4	M1 for factorising $-3x^2 + 12x$ or stating the correct value of b or $b = 3$ embedded in an incorrect final answer in the form $a - 3(x - c)^2$
	$-3\left[\left(x-2\right)^2 .....\right]$ or $-3\left(x-2\right)^2$			M1 for a **correct** first step to complete the square
	$-3\left[\left(x-2\right)^2 - \left(2\right)^2\right]$..... oe or $-3\left(x-2\right)^2 + 12$..... or $-3\left[\left(x-2\right)^2 - \left(2\right)^2 .....\right]$ oe			M1 for a **correct** second step to complete the square
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$29 - 3\left(x-2\right)^2$		A1 oe eg $-3\left(x-2\right)^2 + 29$
				Total 4 marks

25 **ALT**	$-bx^2 + 2bcx - bc^2 + a$ oe or $b = 3$		4	M1 for multiplying out $a - b(x - c)^2$ or stating the correct value of b or $b = 3$ embedded in an incorrect final answer in the form $a - 3(x - c)^2$
	$2bc = 12$ or $a - bc^2 = 17$ oe			M1 for equating coefficients
	2×"3"×c = 12 or a – "3" × "2"^{2} = 17 oe			M1 for finding at least 2 from a or b or c
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	$29 - 3\left(x-2\right)^2$		A1 oe eg $-3\left(x-2\right)^2 + 29$
				Total 4 marks

Answer 32: Nov 2025 P1H Q21 **3 marks**

Topic: Completing the Square Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 33: Nov 2024 P2H Q23 **4 marks**

Topic: Algebraic Proof Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 34: Jun 2024 P2HR Q20 **4 marks**

Topic: Solving Quadratic Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 35: Nov 2023 P2H Q24 **5 marks**

Topic: Solving Quadratic Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 36: Jun 2025 P2HR Q21 **3 marks**

Topic: Solving Inequalities Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 37: Nov 2024 P2H Q20 **3 marks**

Topic: Solving Inequalities Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 38: Nov 2025 P1H Q25 **3 marks**

Topic: Solving Inequalities Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 39: Jun 2022 P2HR Q21 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 40: Jun 2023 P2H Q21 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 41: Jun 2024 P2HR Q22 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 42: Jun 2025 P1H Q22 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 43: Jun 2025 P1HR Q22 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 44: Nov 2024 P2H Q22 **5 marks**

Topic: Simultaneous Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 45: Nov 2025 P2H Q20 **3 marks**

Topic: Forming & Solving Equations Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 46: Jan 2020 P2HR Q22 **4 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 47: Jan 2021 P1H Q24 **4 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 48: Jan 2021 P1HR Q21 **6 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 49: Jan 2022 P2H Q26 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 50: Jun 2022 P1HR Q25 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 51: Jun 2023 P2HR Q23 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 52: Jun 2025 P1H Q24 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 53: Jun 2025 P1HR Q24 **6 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 54: May 2021 P1H Q23 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 55: May 2023 P1H Q20 6 marks

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 56: May 2024 P1H Q24 6 marks

Topic: Sequences Status: matched Source pages: 26,27

24	$\frac{n}{2}\left[2(84)+(n-1)(4)\right]$ or $\frac{n}{2}[168+4n-4]$ or $\frac{n}{2}[164+4n]$ oe or $82n+2n^2$ oe		6	M1 for correctly substituting into $S_n=\frac{n}{2}\left[2a+(n-1)d\right]$
	$\frac{n}{2}\left[2(84)+(n-1)(4)\right]=(n-2)\times 180$ or $\frac{n}{2}[164+4n]=(n-2)\times 180$ oe or $82n+2n^2=(n-2)\times 180$ oe			M1 for equating S_n with $(n-2)180$ S_n must come from correct substitution of a and d into $\frac{n}{2}\left[2a+(n-1)d\right]$
	eg $n^2-49n+180(=0)$ oe Allow $n^2-49n=-180$			M1dep on M2 for multiplying out and collecting terms, forming a three term quadratic in any form of an^2+bn+c (= 0) where at least 2 coefficients (a or b or c) are correct
	eg $(n-45)(n-4)(=0)$ $n=\frac{--49\pm\sqrt{(-49)^2-4\times 1\times 180}}{2}$ e.g. $\left(n-\frac{49}{2}\right)^2-\left(\frac{49}{2}\right)^2=-180$			M1ft dep on M2 method to solve **their** 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as eg $\frac{49\pm\sqrt{2401-720}}{2}$ **or** if factorising allow brackets which expanded give 2 out of 3 terms correct) **or** correct value for $n=45$ (ignore $n=4$)
	("45"$-2)\times 180$ or $\frac{"45"}{2}\left[2(84)+("45"-1)(4)\right]$ oe or ("44"$-2)\times 180$ or $\frac{"44"}{2}\left[2(84)+("44"-1)(4)\right]$ oe			M1 dep on previous M1 NB $n>5$
	Working required	7740		A1 dep on M5 Accept 7560 or 7480
				Total 6 marks

Answer 56 continued

24 ALT	$\frac{n}{2}\left[2(96)+(n-1)(-4)\right]$ or $\frac{n}{2}[192-4n+4]$ or $\frac{n}{2}[196-4n]$ oe or $98n-2n^2$ oe		6	M1 for correctly substituting into $S_n=\frac{n}{2}\left[2a+(n-1)d\right]$ using exterior angles
	$\frac{n}{2}\left[2(96)+(n-1)(-4)\right]=360$ or $\frac{n}{2}[196-4n]=360$ oe or $98n-2n^2=360$ oe			M1 for equating S_n with 360 S_n must come from correct substitution of a and d into $\frac{n}{2}\left[2a+(n-1)d\right]$
	eg $2n^2-98n+360(=0)$ $n^2-49n+180(=0)$ oe Allow $n^2-49n=-180$			M1dep on M2 for multiplying out and collecting terms, forming a three term quadratic in any form of an^2+bn+c (= 0) where at least 2 coefficients (a or b or c) are correct
	e.g $(n-45)(n-4)(=0)$ $n=\frac{--49\pm\sqrt{(-49)^2-4\times1\times180}}{2}$ e.g. $\left(n-\frac{49}{2}\right)^2-\left(\frac{49}{2}\right)^2=-180$			M1ft dep on M2 method to solve **their** 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as eg $\frac{49\pm\sqrt{2401-720}}{2}$ **or** if factorising allow brackets which expanded give 2 out of 3 terms correct) **or** correct value for $n = 45$ (ignore $n = 4$)
	("45"$-2)\times180$ or $\frac{"45"}{2}\left[2(84)+("45"-1)(4)\right]$ oe or ("44"$-2)\times180$ or $\frac{"44"}{2}\left[2(84)+("44"-1)(4)\right]$ oe			M1 dep on previous M1 NB $n > 5$
	Working required	7740		A1 dep on M5 Accept 7560 or 7480
				Total 6 marks

Answer 57: May 2024 P1HR Q23 **4 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 58: May-Nov 2020 P1H Q24 **4 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 59: Nov 2021 P2H Q20 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 60: Nov 2025 P2H Q25 **5 marks**

Topic: Sequences Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 61: Jan 2020 P1H Q21 **5 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 62: Jan 2021 P1H Q22 **3 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 63: Jan 2022 P1HR Q23 6 marks

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 64: Jan 2022 P2H Q22 **5 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 65: Jan 2022 P2H Q25 **5 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 66: Jan 2022 P2HR Q23 **4 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 67: May 2021 P1H Q24 6 marks

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 68: May 2023 P1HR Q21 6 marks

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 69: May-Nov 2020 P1H Q23 **4 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 70: Nov 2023 P1H Q24 **4 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 71: Nov 2024 P1H Q25 **4 marks**

Topic: Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 72: Jan 2020 P1H Q22 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 73: Jan 2020 P2H Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 74: Jan 2020 P2HR Q24 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 75: Jan 2021 P1H Q23 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 76: Jan 2021 P1HR Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 77: Jan 2021 P2HR Q22 **4 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 78: Jan 2022 P1H Q21 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 79: Jan 2022 P1H Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 80: Jan 2022 P1HR Q24 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 81: Jan 2022 P2HR Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 82: Jun 2022 P2H Q22 **7 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 83: Jun 2022 P2H Q23 **2 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 84: Jun 2022 P2HR Q20 **4 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 85: Jun 2023 P2H Q24 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 86: Jun 2023 P2HR Q21 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 87: Jun 2023 P2HR Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 88: Jun 2024 P2H Q22 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 89: Jun 2024 P2H Q25 **2 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 90: Jun 2024 P2HR Q21 **4 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 91: Jun 2025 P1H Q23 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 92: Jun 2025 P1H Q25 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 93: Jun 2025 P2H Q24 **4 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 94: Jun 2025 P2HR Q23 **3 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 95: May 2024 P1HR Q20 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 96: May 2024 P1HR Q22 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 97: May-Nov 2020 P1H Q22 **4 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 98: Nov 2021 P1H Q20 **7 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 99: Nov 2023 P1H Q22 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 100: Nov 2023 P2H Q21 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 101: Nov 2024 P1H Q23 **5 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 102: Nov 2025 P1H Q20 **3 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 103: Nov 2025 P1H Q28 **6 marks**

Topic: Coordinate Geometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 104: Jan 2020 P2HR Q26 **5 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 105: Jan 2021 P1HR Q20 **5 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 106: Jan 2022 P2H Q21 **5 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 107: Jun 2022 P2HR Q25 **5 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 108: Jun 2024 P2H Q24 **6 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 109: Nov 2024 P1H Q24 **6 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 110: Nov 2025 P1H Q27 **6 marks**

Topic: Linear Graphs (y = mx + c) Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 111: Jan 2020 P1H Q20 **4 marks**

Topic: Graphs of Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 112: Jun 2025 P2HR Q25 — 3 marks

Topic: Graphs of Functions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 113: Jun 2022 P1HR Q22 **7 marks**

Topic: Estimating Gradients Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 114: Nov 2024 P1H Q20 4 marks

Topic: Estimating Gradients Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 115: Jan 2020 P1HR Q22 **4 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 116: Jan 2020 P2H Q21 **4 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 117: Jan 2021 P1H Q21 **4 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 118: Jan 2022 P1H Q23 **2 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 119: Jan 2022 P1HR Q20 **6 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 120: Jun 2022 P1H Q20 **2 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 121: Jun 2022 P2HR Q22 **4 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 122: Jun 2024 P2HR Q23 **2 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 123: Jun 2025 P1HR Q21 — 3 marks

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 124: May 2023 P1H Q21 **2 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 125: May 2023 P1HR Q20 **7 marks**

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 126: May 2024 P1H Q21 2 marks

Topic: Transformations of Graphs Status: matched Source pages: 23

21	(i)		(–2, –4)	1	B1
	(ii)		(5, –10)	1	B1
					Total 2 marks

Answer 127: May-Nov 2020 P1H Q25 3 marks

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 128: May-Nov 2020 P1HR Q20 2 marks

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 129: Nov 2021 P2H Q21 6 marks

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 130: Nov 2025 P1H Q24 3 marks

Topic: Transformations of Graphs Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 131: Jan 2020 P1HR Q23 6 marks

Topic: Differentiation Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 132: Jan 2022 P1H Q24 **6 marks**

Topic: Differentiation Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 133: Jun 2025 P2HR Q22 **5 marks**

Topic: Differentiation Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 134: May 2021 P2H Q21 **5 marks**

Topic: Differentiation Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 135: Jan 2022 P2HR Q21 **6 marks**

Topic: Standard & Compound Units Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 136: Jun 2025 P1H Q20 **5 marks**

Topic: Standard & Compound Units Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 137: Nov 2025 P2H Q26 **6 marks**

Topic: Standard & Compound Units Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 138: Jan 2021 P2H Q20 **4 marks**

Topic: Angles in Polygons & Parallel Lines Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 139: Jun 2022 P1H Q23 **6 marks**

Topic: Angles in Polygons & Parallel Lines Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 140: May 2023 P1H Q22 **6 marks**

Topic: Angles in Polygons & Parallel Lines Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 141: Jun 2023 P2HR Q25 6 marks

Topic: Bearings, Scale Drawing & Constructions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 142: Jun 2025 P2HR Q24 **6 marks**

Topic: Bearings, Scale Drawing & Constructions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 143: Nov 2024 P2H Q24 6 marks

Topic: Bearings, Scale Drawing & Constructions Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 144: Jun 2023 P2HR Q20 **2 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 145: Jun 2024 P2H Q21 **5 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 146: Jun 2025 P1HR Q23 **5 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 147: Nov 2023 P2H Q20 **4 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 148: Nov 2024 P2H Q25 **5 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 149: Nov 2025 P2H Q22 **6 marks**

Topic: Circle Theorems Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 150: Jan 2021 P2HR Q20 **5 marks**

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 151: Jun 2022 P2H Q20 **4 marks**

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 152: Jun 2023 P2H Q22 **5 marks**

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 153: May 2021 P1H Q25 **6 marks**

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 154: May 2021 P1H Q26 **5 marks**

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 155: May 2023 P1HR Q25 6 marks

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 156: May-Nov 2020 P1HR Q22 6 marks

Topic: Area & Perimeter Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 157: Jan 2020 P2H Q26 **6 marks**

Topic: Circles, Arcs & Sectors Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 158: Jan 2021 P2H Q22 **6 marks**

Topic: Circles, Arcs & Sectors Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 159: Jun 2025 P1HR Q25 **6 marks**

Topic: Circles, Arcs & Sectors Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 160: May 2024 P1H Q20 4 marks

Topic: Circles, Arcs & Sectors Status: matched Source pages: 22

20	eg $\pi r^2 \times \frac{60}{360} - \frac{1}{2} r^2 \sin 60$ oe or $\frac{\pi r^2}{6} - \frac{\sqrt{3}}{4} r^2$ oe		4	M1 for a correct expression for the area of the segment Expression may be embedded in an equation, eg $\pi r^2 \times \frac{60}{360} - \frac{1}{2} r^2 \sin 60 = 38$ or $\pi r^2 \times \frac{60}{360} = 38 + \frac{1}{2} r^2 \sin 60$ or $\pi r^2 \times \frac{60}{360} - 38 = \frac{1}{2} r^2 \sin 60$
	eg $\left(r^2 =\right) 38 \div \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right) \left(= 38 \div 0.09(058)\right)$ (= 419(.490…)) oe or $\left(r =\right) \sqrt{38 \div \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)}$ (= 20.4(81…))oe			M1 dep on M1 for a **correct** expression for r^2 or r
	$\frac{\pi}{6} \times "20.4(81...)" \times 2$ (= 21.4(48…)) oe or			M1 for using the value of r to find arc length
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	41.9		A1 allow 41 - 42
				Total 4 marks

Answer 161: Jun 2022 P1H Q22 **5 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 162: Jun 2023 P2HR Q24 **5 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 163: Jun 2025 P2H Q21 **3 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 164: Jun 2025 P2HR Q26 **5 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 165: May 2024 P1H Q23 5 marks

Topic: Volume & Surface Area Status: matched Source pages: 25

23	$\frac{1}{3}\pi\times(5x)^2\times 6x$ oe or $50\pi x^3$ oe or $\frac{1}{2}\times\frac{4}{3}\times\pi\times(2x)^3$ or $\frac{16}{3}\pi x^3$ oe or $\frac{4}{3}\times\pi\times(2x)^3$ or $\frac{32}{3}\pi x^3$ oe		5	M1 for finding the volume of cone or hemisphere or sphere NB Ignore missing brackets around $5x$ and $2x$ for this mark
	$\frac{1}{3}\pi\times(5x)^2\times 6x-\frac{1}{2}\times\frac{4}{3}\times\pi\times(2x)^3=6948\pi$ oe or $50\pi x^3-\frac{16}{3}\pi x^3=6948\pi$ or $\frac{134}{3}\pi x^3=6948\pi$ oe			M1 for a **correct** equation for the volume of the shape NB If not expanded at this stage then must see brackets
	$\left(x^3=\right)\frac{6948\pi\times 3}{134\pi}\left(=\frac{10422}{67}=155.(552...)\right)$ oe or $\left(x=\right)\sqrt[3]{\frac{6948\pi\times 3}{134\pi}}\left(=\sqrt[3]{\frac{10422}{67}}=\sqrt[3]{155.(552...)}=5.37(8...)\right)$ oe			M1 for rearranging the **correct** equation to find the value of x^3 or x Accept 5.4 or better
	$3\times\pi\times(2\times"5.37(8...)")^2$ oe or $12\times\pi\times$ "$5.37(8..)$"^{2} oe			M1 for finding the surface area of the hemisphere
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	1090		A1 allow 1086 – 1100 Special case for using 6948 without π SC B3 for $x^3 = 49.5(138\ldots)$ or $x = 3.67(205\ldots)$ SC B4 for awrt 508
				Total 5 marks

Answer 166: Nov 2023 P1H Q21 **5 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 167: Nov 2024 P1H Q22 **3 marks**

Topic: Volume & Surface Area Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 168: Jan 2020 P1HR Q20 **5 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 169: Jan 2020 P2HR Q20 **4 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 170: Jan 2021 P1H Q20 **3 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 171: Jun 2022 P2HR Q24 **6 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 172: Jun 2024 P2HR Q24 **5 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 173: Jun 2025 P2H Q26 **4 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 174: May 2021 P2H Q20 **3 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 175: May 2023 P1H Q23 **5 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 176: May-Nov 2020 P1H Q20 3 marks

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 177: Nov 2023 P1H Q23 **5 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 178: Nov 2023 P2H Q23 **5 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 179: Nov 2025 P2H Q23 **3 marks**

Topic: Area & Volume of Similar Shapes Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 180: Jan 2020 P2H Q20 **3 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 181: Jan 2020 P2H Q25 3 marks

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 182: Jan 2020 P2HR Q25 **3 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 183: Jan 2021 P2HR Q21 **6 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 184: Jun 2023 P2H Q23 **4 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 185: Nov 2021 P2H Q22 **6 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 186: Nov 2025 P1H Q26 **5 marks**

Topic: Right-Angled Triangles - Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 187: Jan 2020 P1HR Q21 **6 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 188: Jan 2020 P2H Q24 **4 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 189: Jan 2021 P2H Q21 **4 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 190: Jan 2022 P2H Q23 **6 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 191: Jun 2022 P2HR Q23 **3 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 192: Jun 2023 P2HR Q26 **3 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 193: Jun 2024 P2HR Q26 **4 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 194: May 2021 P1H Q21 **4 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 195: May-Nov 2020 P1H Q21 **5 marks**

Topic: Sine, Cosine Rule & Area of Triangles Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 196: Jun 2025 P1HR Q20 3 marks

Topic: 3D Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 197: May 2021 P1H Q22 **4 marks**

Topic: 3D Pythagoras & Trigonometry Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 198: Jun 2024 P2HR Q25 **6 marks**

Topic: Vectors Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 199: Nov 2023 P1H Q20 **4 marks**

Topic: Statistics Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 200: Jan 2022 P1HR Q21 3 marks

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 201: Jan 2022 P2H Q20 4 marks

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 202: Jun 2022 P1H Q21 3 marks

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 203: Jun 2025 P2HR Q20 **4 marks**

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 204: May 2023 P1HR Q22 **4 marks**

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 205: May 2024 P1H Q22 3 marks

Topic: Histograms Status: matched Source pages: 24

22	eg 14 ÷ 5 (= 2.8) or a correct value on the FD scale or 10 small squares =1 adult oe or 1 large square = 2.5 adults oe or 51 **and** 8 assigned to correct bars (distances)		3	M1 for finding the frequency density **or** for finding the number of adults for squares **or** use of counting squares or blocks
	eg 14 + (15 × "3.4") + (20 × "0.4") (= 73) oe or 100 – [14 + (15 × "3.4") + (20 × "0.4")] (= 27) oe 14 + 51 + 8 (= 73) oe or 100 – [14 + 51 + 8] (= 27) oe or (140 + 510 + 80) × 0.1 (= 73) oe or [1000 – (140 + 510 + 80)] × 0.1 (= 27) oe or (140 + 510 + 80) (= 730) oe or [1000 – (140 + 510 + 80)] (= 270) oe or (5.6 + 20.4 + 3.2) × 2.5 (= 73) oe or [40 – (5.6 + 20.4 + 3.2)] × 2.5 (= 27) oe			M1 for a method to find the area of the bars given or for a method to find the missing area
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	Correct height of bar at 2.7 and correct width		A1 for correct bar(s) with frequency of 27 SC B2 for a bar of height 2.7 from 0 – 15 SC B2 for a bar of height 1.8 from 0 – 15
				Total 3 marks

Answer 206: Nov 2025 P2H Q21 3 marks

Topic: Histograms Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 207: Jun 2022 P1HR Q24 **5 marks**

Topic: Probability Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 208: Jun 2025 P1H Q21 3 marks

Topic: Probability Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 209: May 2021 P2H Q23 **5 marks**

Topic: Probability Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 210: May 2024 P1HR Q21 **5 marks**

Topic: Probability Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 211: May-Nov 2020 P1HR Q23 4 marks

Topic: Probability Toolkit Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 212: Jun 2025 P2H Q20 **6 marks**

Topic: Probability Diagrams - Venn & Tree Diagrams Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 213: Jun 2023 P2H Q20 **3 marks**

Topic: Combined & Conditional Probability Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.

Answer 214: Nov 2025 P1H Q22 3 marks

Topic: Combined & Conditional Probability Status: missing_source Source pages: -

Missing Answer Source

Add a mark scheme or worked-solution PDF to answer_sources_manifest.csv

Add or correct the source in `answer_sources_manifest.csv`, then rerun the script.